



ACL 2026 FINDINGS

# PERSONAFORGE

Jizhou Tong (Wuhan University) & Sirui Zou (SWUFE)

**OPEN SOURCE**  
Please give us a star!



## ABSTRACT

Large Language Models excel at role-playing but struggle to maintain consistent personalities across extended multi-turn interactions. PersonaForge introduces a **Three-Layer Personality Architecture** combined with a **Dual-Process Generation Mechanism**.

- **Orthogonality:** Big Five + Defense Mechanisms reduce long-dialogue drift.
- **Integration:** High-dimensional constraints require an Inner Monologue workspace to resolve.

## ARCHITECTURE

1. **Core Traits:** Big Five scores (O,C,E,A,N) + Vaillant's Defense Mechanisms.
2. **Style Layer:** Constraints on sentence length, register, punctuation.
3. **Dynamic State:** Tracks mood, energy, and relationship intimacy.

[ USER MESSAGE ]



**Trigger Detector**  
(Selective)



**Inner Monologue**  
(System 2)



**Styled Response**  
(System 1)

PERSONALITY CONSISTENCY  
(PC)

**+19.4%**  
Over Structured-CoT Baseline

CHARACTER DRIFT RATE

**6.3%**  
Reduced from 24.8% (50-Turn Eval)

DEFENSE MECH ACTIVATION

**+64.7%**  
Psychological Depth Reconstructed

## KEY INSIGHTS

**Catastrophic Rigidity in SFT:** Low-resource SFT learns "how to speak" but not "who I am", leading to 57.7% repetition and 66.0% drift.

**Dual-Process Efficiency:** Think-then-Speak activates only for critical interactions (~40%), achieving 96% performance at 13.4% overhead.

## ROBUSTNESS & VALIDATION

**Adversarial Defense:** Maintained character consistency in **95.2%** of 21-turn RoleBreak attacks.

